



Hyundai's robotic assistant

The South Korean auto-maker is working on wearable robotics using smart exoskeleton assistants. <http://dgit.in/HyundaiR>



Improved eyecare

Microsoft Intelligent Network for Eyecare plans to use machine learning to improve eyecare in India. <http://dgit.in/MSeyeIND>

Space Age

contribute to gamma outbursts. This one is fairly similar to the SN2005gl supernova observed in 2005, and the more recent, still-under-debacle ASASSN-15lh explosion. The reason for the formation of superluminous energies here is the consistent observation of how these stars are embedded within corresponding nebulae - a highly proactive galactic



When birthplace meets death

zone rich in circumstellar material. These are the elements that give birth to a star, and upon its death, the energy jets react with the matter to create visible shockwaves of stellar energy. Interestingly, the formation of superluminous supernovae from circumstellar matter do not necessarily depend on the constituency of the star, but more on its spatial arrangement.

There are other observed processes in the formation of massive supernovae, as well, but they get increasingly rarer. For instance, super-massive stars in the order of 260 times the mass of Sun give rise to particles and antiparticles upon explosion. The existence of opposing matter leads to extremely rapid partial collapse of the core, which further causes simultaneous fusion of matter aided by gravity, all of which leads to the complete death of a star, and an extreme magnitude of energy and light repulsion that can be as much as one hundred times more intense than the highest standard supernova. The process has not yet been confirmed in interstellar space, seeing that the sheer size of stars that can have such reactions are not present in our immediate universe.

The birth of a magnetar, as explained by Matthew Smith, post-doctoral re-

searcher at the University of Southampton to Phys.org, can also be a plausible reason behind the birth of a superluminous supernova. The phenomenon would involve the birth of a rapidly spinning neutron star, preceded by extreme heat and energy followed by rapid cooling. While the rapid cooling observation would be somewhat against a superlu-



High intensity deaths can be titanic in magnitude

minous supernova, the observations are so far being studied to collate further information, before inferring.

Probability and geometry

The core collapse model also goes beyond probability factors of understanding superluminous supernovae. Cosimo Inserra and his team from Queens University, Belfast, have obtained spectropolarimetric data on recent hypernova observations to affirm that despite an asymmetric, spherical pattern of the stars in explosion, the formation pivots around a primary axis, and the polarisation of matter changes in time and wavelength.

This coincides with the core-collapse model of a star's death, further proving that while superluminous supernovae have been much recent discoveries, they have actually been an ancient phenomena, and may not even be as rare going beyond our present, engineering limits.

A new milestone for space research

While not much have been attributed to the impact of superluminous supernovae in terms of terrestrial impact, they do contribute further to the study of black holes, neutron stars and the core

reactions in massive stars. It also lends basis to the study of gamma radiation bursts, which till the turn of the millennium was a matter completely shrouded in mystery. Even with the plausible explanations, the matter of short-duration gamma bursts still remain in question, and the hypernovae only contribute to long-duration bursts.

While hyper-size star theories still remain in question because of the rarity, a possible massive gamma radiation burst has already been attributed to the second-largest extinction ever on Earth, during the Hirnantian Age of the Ordovician Period, more than 440 million years ago. While this was only ever a hypothesis and no concrete evidence was found, it may have been on the impact of a massive star's violent death due to matter-antimatter collision.

Our immediate future seems to be more secure, and mankind has far greater threat of extinction from global climate change than from an imminent hypernova. With every discovery we escalate to a new height of stellar explosion force that exceeds previous gargantuan figures significantly. The ongoing research on superluminous supernovae is a major milestone for



Now this won't ever happen... Hopefully

space research, because of these very titanic forces at play. These aptly define the endless borders of space, and how practically infinite can the titanic nature of forces be.

It is occurrences like these that show you the true power of nature. On the other hand, it is truly a sight to behold. And maybe someday we will find the fourth type of supernova that's even more awesome. How big would that be? Let us know through your messages 